# AutomationEdge

## NPCI UPI Reconciliation – Banker's Master Guide & Demo Cheat Sheet

50,000 RECORDS / BATCH
✓ RBI T+1 REGULATORY READY
Unified Operations Workbench • Multi-Leg In-Memory Engine

## 1. Executive Overview & The Banking Challenge

In Indian retail digital payments, **UPI (Unified Payments Interface)** processes over 400 million transactions daily. A single transaction travels through 4 distinct software systems: the national clearing rail (NPCI), the bank's transaction gateway switch (Base24), the Core Banking System customer ledger (Finacle), and internal inter-bank accounting suspense accounts (GL-14020). Because of unpredictable network latencies, timeouts, and system drifts, data breaks routinely happen.

**The Operational Challenge for Banks:**
Manual or batch-based recon cycles take between **6 to 18 hours**. When customer debits fail without auto-reversal, banks violate the **RBI T+1 Harmonization Mandate**, incurring a non-negotiable statutory penalty of ■**100 per day** payable directly into the affected customer's account.

**The AutomationEdge Solution:**
Autonomous 4-way in-memory matching of **50,000 transactions in under 2 seconds**, achieving **99.95% Straight-Through Processing (STP)**, triggering instantaneous customer account refunds for deemed failures, and compiling cryptographic audit dossiers for high-risk exceptions.

## 2. Banking Terms & Acronyms Glossary

Standard terms used in retail payments, core banking, settlement, and reconciliation:

| Term | Full Form | Definition & Role in Reconciliation |
|---|---|---|
| **NPCI** | National Payments Corporation of India | Central umbrella entity governed by RBI operating UPI, IMPS, NACH, and RuPay rails. |
| **STP** | Straight-Through Processing | Transactions that reconcile and clear 100% automatically without human touch. Target: >99.9%. |
| **HITL** | Human-in-the-Loop | Maker-checker supervisory interface where high-value breaks (> ■10k) require operations approval. |
| **CBS** | Core Banking System | Central software maintaining customer deposit/loan ledgers (e.g., Infosys Finacle, TCS BaNCS). |
| **Base24** | ACI Base24 Payment Switch | Bank gateway switch routing live transaction traffic (00=Auth, 91=Timeout, 96=Error). |
| **GL Suspense** | General Ledger (GL-14020) | Inter-bank parking pool holding funds during the clearing cycle. Must liquidate to ■0.00 variance. |
| **RRN** | Retrieval Reference Number | Unique 12-digit numeric identifier generated by the switch. Primary correlation key across all 4 files. |
| **UTR** | Unique Transaction Reference | Alphanumeric transaction identifier generated for inter-bank clearing. Secondary matching key. |
| **P2P** | Peer-to-Peer (Person-to-Person) | Direct account transfer between two individuals (e.g., Google Pay / PhonePe friend transfer). |
| **P2M** | Peer-to-Merchant | Payment from an individual to a registered shop or business (QR code scanning at retail outlets). |
| **Deemed Fail** | Deemed Failed Transaction | Customer account debited, but switch timed out or beneficiary failed. Mandated T+1 auto-refund. |
| **Zero-Delta** | Zero Variance Balance | State where Total Debits equal Total Credits in suspense, leaving exactly ■0.00 difference. |
| **Knock-Off** | Ledger Liquidation | Posting balancing entries to clear matched items out of suspense into settlement accounts. |

## 3. The 4 Inbound Settlement Feeds: Source & Direction

Reconciliation happens **inside the Bank**. To ensure end-to-end auditability, the bank correlates **1 external file received from NPCI** against **3 internal bank systems**:

| File Name | Source & Direction | Schema Fields | Operational Role in Bank |
|---|---|---|---|
| **NPCI_UPI_Settlement_Master.csv** | **NPCI ➔ Bank** (External via SFTP) | `RRN, TXN_ID, UTR, RAIL, CYCLE_ID, AMOUNT, STATUS` | Official clearing file pushed by NPCI at the end of each cycle. Central source of network settlement truth. |
| **Switch_Base24_Logs.csv** | **Bank Payment Switch** (Internal Gateway) | `RRN, SWITCH_STATUS, SWITCH_RESP_CODE` | Real-time authorization states captured at the bank's gateway (00=Auth, 91=Timeout, 96=Error). |
| **Finacle_CBS_Dump.csv** | **Core Banking System** (Internal Finacle) | `RRN, CBS_AMOUNT, CBS_STATUS, ACCOUNT_NO` | Bank customer deposit ledger showing if the customer account was actually debited or credited. |
| **GL_Suspense_Pool.csv** | **General Ledger** (Internal Accounting) | `RRN, GL_AMOUNT, GL_STATUS` | Inter-bank settlement suspense balance (GL-14020). Must be knocked off to zero delta. |

## 4. 4-Way Matching Logic & Decision Matrix

Every transaction record is joined on RRN and UTR across all 4 legs, and classified into 4 precise action paths:

| Category | Condition Across 4 Legs | Automated Action | Business Result |
|---|---|---|---|
| **1. 4-Way STP Match (99.95% volume)** | • NPCI = SUCCESS<br>• Switch = 00 (Approved)<br>• CBS = DEBITED<br>• GL = PARKED<br>• All Amounts Equal | Dispatched to **Zero-Delta GL Knock-Off** workflow (UPI_ZeroDelta_GL_KnockOff.psw) | Balances GL-14020 & GL-11010. **■0.00 ledger variance**. Instant cycle close. |
| **2. Deemed Failure Auto-Refund (Break < ■5,000)** | • Switch = 91 (Timeout) / 96<br>• CBS = DEBITED (Customer charged)<br>• NPCI = Failed / Absent<br>• Amount < ■5,000 | Dispatched to **CBS Auto-Refund Execution** workflow (UPI_CBS_Execution_Bot.psw) | Calls Finacle API to credit customer account. **100% compliance with RBI T+1 TAT** (stops penalty). |
| **3. High Value / Mismatch Break (> ■10k or Variance)** | • Amount mismatch across legs, OR<br>• High-value exception > ■10,000 | Dispatched to **AI Exception & HITL Dispatcher** (UPI_AI_DualTrack_Dispatcher.psw) | Compiles multi-system dossier with 94% AI confidence score for 1-click maker-checker signoff. |
| **4. Orphan GL Balance** | • Balance parked in GL-14020<br>• No matching CBS debit or NPCI clearance | Flagged for Treasury Clearance | Prevents unbacked balances from rolling over into next day's settlement. |

## 5. Regulatory Compliance: The RBI T+1 Mandate

**RBI Circular Reference: DPSS.CO.PD No.629/02.01.014/2019-20**
*'Harmonization of Turn Around Time (TAT) and customer compensation for failed transactions'*

• **Mandated Turnaround:** For failed UPI transactions where the customer account was debited, the customer must receive credit reversal within **T + 1 calendar day**.
• **Statutory Penalty:** If credit is delayed beyond T+1, the bank is legally mandated to compensate the customer at **■100 per day of delay** automatically without waiting for a customer dispute.
• **AutomationEdge Protection:** By identifying deemed failures at file ingest time and executing CBS reversal workflows in under 2 seconds, the bank completes the reversal well within T+1, resulting in **■0.00 penalty exposure**.

## 6. AutomationEdge Execution Tools (Process Studio Workflows)

The Reconciler Agent dispatches batches directly into 3 specialized Process Studio execution tools via REST API:

| Tool Name | Workflow File | Trigger Condition | Action & Operational Outcome |
|---|---|---|---|
| **Zero-Delta GL Knock-Off** | `UPI_ZeroDelta_GL_KnockOff.psw` | Trigger: STP Matched (99.95%) | Auto-posts balancing journal entry between GL-14020 & GL-11010. ■**0.00 ledger variance** and instant cycle closure. |
| **CBS Auto-Refund Execution** | `UPI_CBS_Execution_Bot.psw` | Trigger: Auto-Refund < ■5,000 | Calls Finacle CBS reversal API and credits affected customer account. **100% compliance with RBI T+1 TAT** (stops penalty clock). |
| **AI Exception & HITL Dispatcher** | `UPI_AI_DualTrack_Dispatcher.psw` | Trigger: High Value (>■10k) / Mismatches | Compiles multi-system evidence dossiers with 94% AI confidence score for 1-click maker-checker signoff. |

## 7. Demo Script & Anticipated Banker Q&A;

**■■ 30-Second Elevator Pitch for Tomorrow's Presentation:**
*'In day-to-day operations, bank reconciliation teams spend 6 to 18 hours manually matching raw NPCI settlement files against Base24 switch logs, Finacle customer accounts, and GL suspense pools to detect breaks.*

*AutomationEdge ingests all 4 settlement feeds, correlates 50,000 records in less than 2 seconds, automatically closes out the 99.95% straight-through matches to zero-delta balance, and triggers instant CBS auto-refunds for deemed failures—ensuring 100% compliance with the RBI T+1 circular with zero human intervention.'*

**Top Anticipated Questions from Bank Employees:**

**Q: Why does our dataset have exactly 25 breaks in 50,000 records?**

A: In real-world Indian banking production, UPI STP rates average between 99.9% and 99.95%. Having 25 breaks out of 50,000 accurately mirrors authentic bank volumes—proving both high-speed automated throughput and intelligent edge-case routing.

**Q: How does AutomationEdge connect to Finacle CBS and Base24?**

A: Through secure REST APIs, direct database connectors, or scheduled batch SFTP drops. Process Studio workflows can call Finacle reversal APIs directly or generate pre-formatted journal upload files.

**Q: What happens if an amount mismatch is found? Does it auto-refund?**

A: No. Any transaction with differing amounts across legs is halted immediately from automated posting. It is routed to the Human-in-the-Loop Cockpit with an AI evidence dossier showing the exact rupee divergence for maker-checker signoff.

**Q: Can statutory and RBI auditors verify the automated decisions?**

A: Yes. Every single transaction matching step, decision rule, and reversal execution generates an immutable JSON audit trail stamped with SHA-256 cryptographic verification for statutory audit compliance.